

## Calculus I Lesson 45

1) Use a calculator and find  $\log_2 \frac{1}{8} = \underline{\hspace{2cm}} ?$

Hint:  $\log_2 \frac{1}{8} = \frac{\ln(1/8)}{\ln(2)}$

2) Use a calculator and find  $\log_4 1 = \underline{\hspace{2cm}} ?$

Hint:  $\log_4 1 = \frac{\ln(1)}{\ln(4)}$

3) Solve the log equation  $\log_{10} x = 2$ .

Hint:  $\log_b z = t \iff z = b^t$ .

$x = \underline{\hspace{2cm}} ?$

4) Solve the log equation  $\log_4 x = 3$ .

Hint:  $\log_b z = t \Leftrightarrow z = b^t$ .

$$x = \underline{\hspace{1cm} ? \hspace{1cm}}$$

5) Solve the log equation  $\log_3 (x - 4) = 3$ .

Hint:  $\log_b z = t \Leftrightarrow z = b^t$ .

$$x = \underline{\hspace{1cm} ? \hspace{1cm}}$$

6) Solve the log equation  $\log_5(3x - 14) = 4$ .

Hint:  $\log_b z = t \Leftrightarrow z = b^t$ .

$$x = \underline{\hspace{1cm} ? \hspace{1cm}}$$

7)  $f(x) = \log_4(5x + 11)$ .

Find  $f'(x) = \underline{\hspace{1cm} ? \hspace{1cm}}$

Hint:  $D_x(\log_a u) = \frac{1}{\ln a} \cdot \frac{1}{u} \cdot \frac{du}{dx}$       Using Chain Rule

a) Let  $u = 5x + 11$ .       $\frac{du}{dx} = \underline{\hspace{1cm} ? \hspace{1cm}}$

b)  $D_x(\log_4(5x + 11)) = \underline{\hspace{1cm} ? \hspace{1cm}}$

8)  $f(x) = \log_5(4 - 5x)^2$ .

Find  $f'(x) = \underline{\hspace{2cm}}?$

Hint:  $D_x(\log_a u) = \frac{1}{\ln a} \cdot \frac{1}{u} \cdot \frac{du}{dx}$       Using Chain Rule

Hint:  $\log_5(4 - 5x)^2 = 2 \cdot \log_5(4 - 5x)$ .

a) Let  $u = 4 - 5x$ .       $\frac{du}{dx} = \underline{\hspace{2cm}}?$

b)  $D_x[(2 \cdot \log_5(4 - 5x))] = \underline{\hspace{2cm}}?$

9)  $f(x) = \log_5 \sqrt{x^2 - 5}$ .

Find  $f'(x) = \underline{\hspace{2cm}}?$

Hint:  $D_x(\log_a u) = \frac{1}{\ln a} \cdot \frac{1}{u} \cdot \frac{du}{dx}$       Using Chain Rule

Hint:  $\log_5 \sqrt{x^2 - 5} = \log_5 (x^2 - 5)^{1/2} = \frac{1}{2} \cdot \log_5 (x^2 - 5)$ .

a) Let  $u = x^2 - 5$ .       $\frac{du}{dx} = \underline{\hspace{2cm}}?$

b)  $D_x \left[ \frac{1}{2} \cdot \log_5 (x^2 - 5) \right] = \underline{\hspace{2cm}}?$





